

Attention Is All You Need

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, Illia Polosukhin

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The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely.

Introduction

Recurrent neural networks, long short-term memory and gated recurrent neural networks in particular, have been firmly established as state of the art approaches in sequence modeling and transduction problems such as language modeling and machine translation. Recurrent models typically factor computation along the symbol positions of the input and output sequences. This inherently sequential nature precludes parallelization within training examples, which becomes critical at longer sequence lengths.

Methodology

The Transformer follows this overall architecture using stacked self-attention and point-wise, fully connected layers for both the encoder and decoder. An attention function can be described as mapping a query and a set of key-value pairs to an output, where the query, keys, values, and output are all vectors. The output is computed as a weighted sum of the values, where the weight assigned to each value is computed by a compatibility function of the query with the corresponding key. Multi-head attention allows the model to jointly attend to information from different representation subspaces at different positions.

Experiments & Evaluation

We trained on the standard WMT 2014 English-German dataset consisting of about 4.5 million sentence pairs. Sentences were encoded using byte-pair encoding. For English-French, we used the significantly larger WMT 2014 English-French dataset consisting of 36M sentences. On the WMT 2014 English-to-German translation task, the big transformer model achieves a state-of-the-art BLEU score of 28.4.

Results & Discussion

The Transformer achieves state-of-the-art translation quality while being more parallelizable and requiring significantly less time to train. Table 2 summarizes our results. On the English-to-German task, our model outperforms the previously best reported results (including ensembles) by more than 2.0 BLEU. Training took 3.5 days on 8 P100 GPUs.

Limitations

While the Transformer eliminates recurrence, self-attention exhibits quadratic computational and memory complexity $O(N^2)$ with respect to sequence length N . This poses significant challenges when processing long context documents, high-resolution imagery, or audio sequences without sparse or hierarchical approximations.

Conclusion & Future Work

In this work, we presented the Transformer, the first sequence transduction model based entirely on attention, replacing the recurrent layers most commonly used in encoder-decoder architectures with multi-headed self-attention. We plan to extend the Transformer to problems involving input and output modalities other than text.